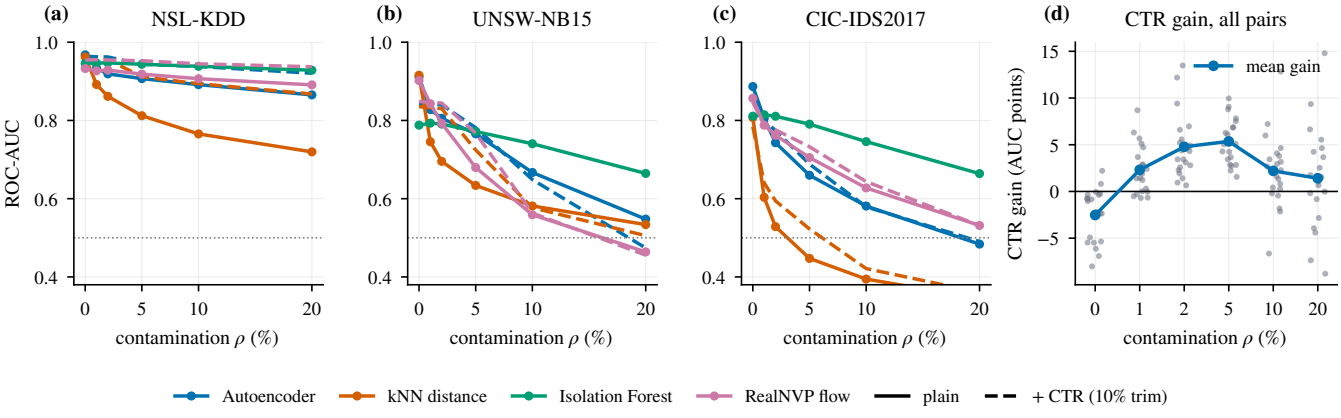
(a)
NSL-KDD
(b)
UNSW-NB15
(c)
CIC-IDS2017
(d)
CTR gain, all pairs
ROC-AUC
1.0
0.8
0.6
0.4
0
5
10
20
contamination ρ (%)
CTR gain (AUC points)
15
10
5
0
−5
1
2
mean gain
Autoencoder
kNN distance
Isolation Forest
RealNVP flow
plain
+ CTR (10% trim)